EASY · ADVANCED MATH

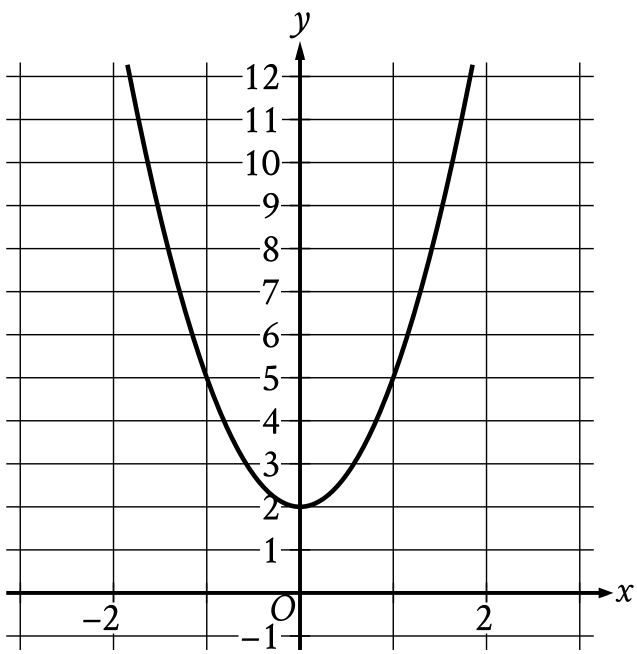
Nonlinear functions

04

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29



- The parabola opens upward.
- The parabola passes through the following points:
  - (negative 1 comma 5)
  - (0 comma 2)
  - (1 comma 5)

The graph of the quadratic function $y = fx$ is shown. What is the vertex of the graph?

A. $0, -2$

B. $0, -3$

C. $0, 2$

D. $0, 3$

Q2

ADVANCED MATH

The function f is defined by $f\left(x\right) = 4 + \sqrt{x}$. What is the value of $f 144$?

A. 0

B. 16

C. 40

D. 76

Q3 ADVANCED MATH

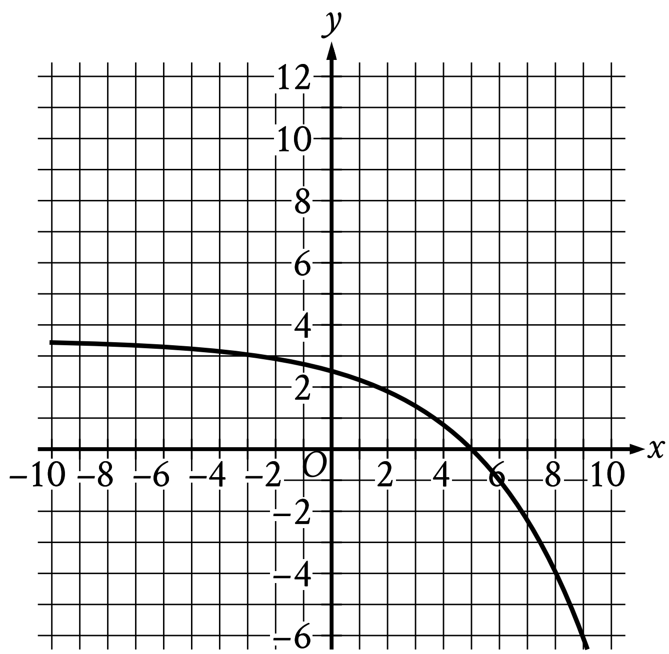
The function f is defined by $f x = x^3 + 15$. What is the value of $f 2$?

A. 20

B. 21

C. 23

D. 24



- Moving from left to right:
    - The curve passes from quadrant 2 to quadrant 1 to quadrant 4.
    - In quadrant 2, the curve trends down gradually to the approximate point (0 comma 2.5).
    - In quadrant 1, the curve trends down sharply to point (5 comma 0).
    - In quadrant 4, the curve trends down sharply.
- The curve passes through the following points:
    - approximately (0 comma 2.5)
    - approximately (4 comma 0.8)

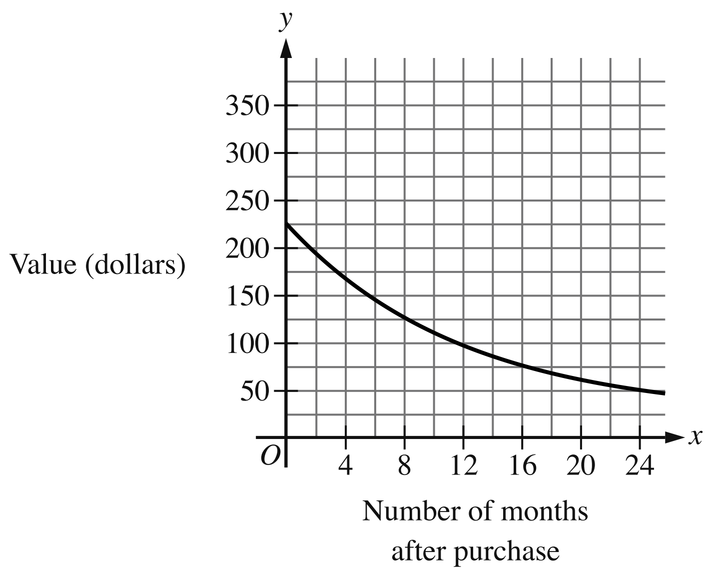
What is the *x*-intercept of the graph shown?

A. -5, 0

B. 5, 0

C. $-2, 0$

D. $2, 0$



- Moving from left to right:
    - The curve is in quadrant 1.
    - The curve trends down gradually.
- The curve passes through the following approximate points:
    - (0 comma 225)
    - (24 comma 50)

The graph shown gives the estimated value, in dollars, of a tablet as a function of the number of months since it was purchased. What is the best interpretation of the y-intercept of the graph in this context?

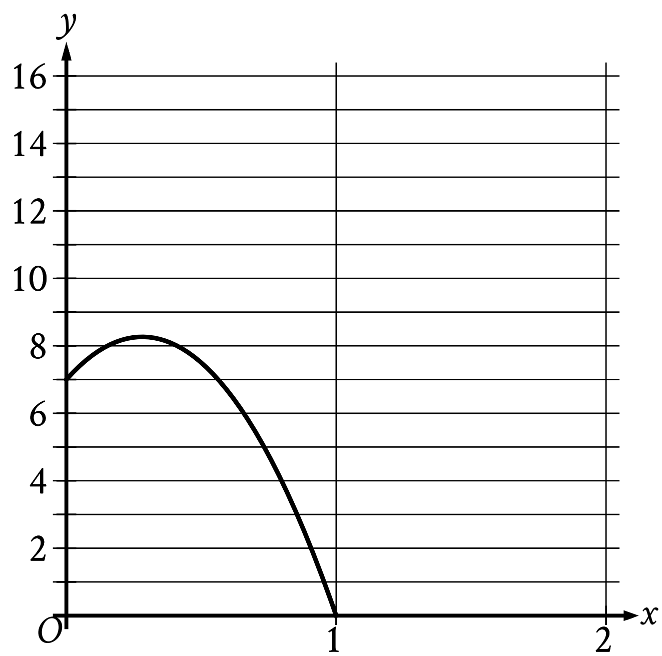
A. The estimated value of the tablet was $\$225$ when it was purchased.

B. The estimated value of the tablet 24 months after it was purchased was $\$225$.

C. The estimated value of the tablet had decreased by $\$225$ in the 24 months after it was purchased.

D. The estimated value of the tablet decreased by approximately 2.25% each year after it was purchased.

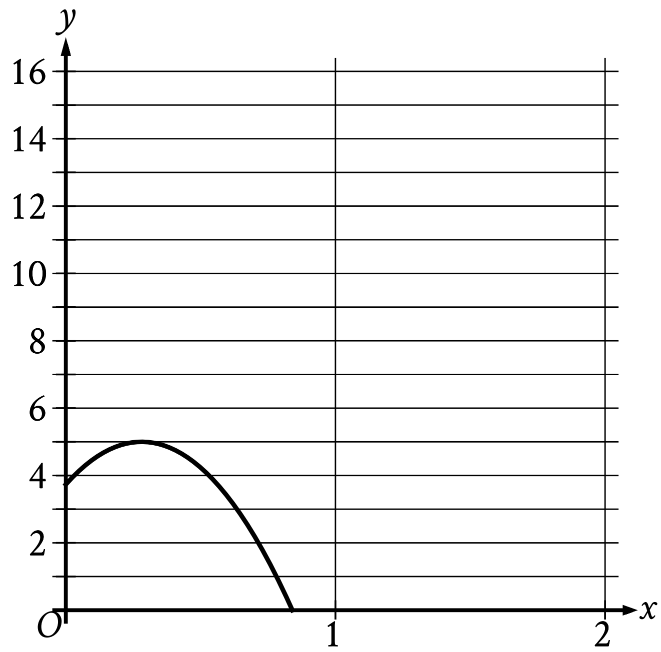
During the first part of an experiment, a ball was launched from a 7-foot-tall platform. The graph shows the height y, in feet, of the ball x seconds after it was launched during the first part of the experiment.



- The parabola opens downward.
- The vertex is at the approximate point (0.3 comma 8.3).
- The parabola passes through the following points:
  - (0 comma 7)
  - approximately (0.3 comma 8.3)
  - approximately (0.6 comma 7)

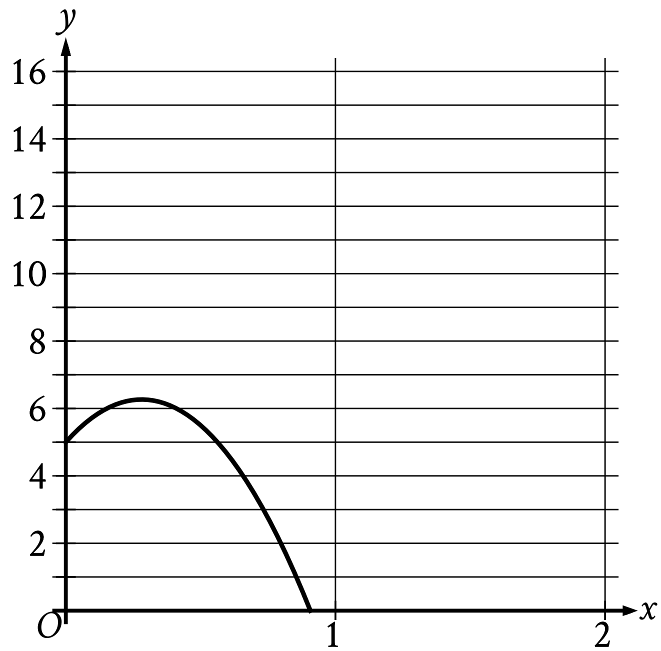
During the second part of the experiment, the ball was launched the same way, but from a platform that is 2 feet shorter than the first platform. Which of the following graphs could represent the height y, in feet, of the ball x seconds after it was launched during the second part of the experiment?

A.



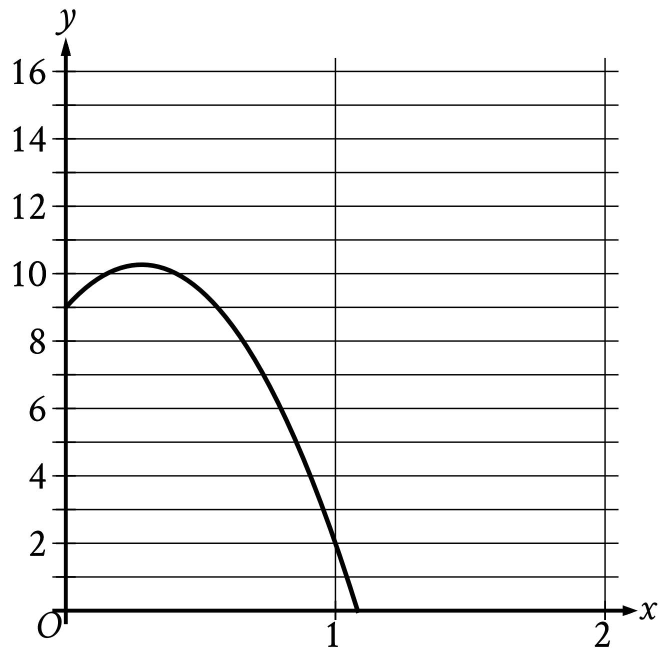
- The parabola opens downward.
- The vertex is at the approximate point (0.3 comma 5.0).
- The parabola passes through the following points:
  - approximately (0 comma 3.7)
  - approximately (0.3 comma 5.0)
  - approximately (0.6 comma 3.7)

B.



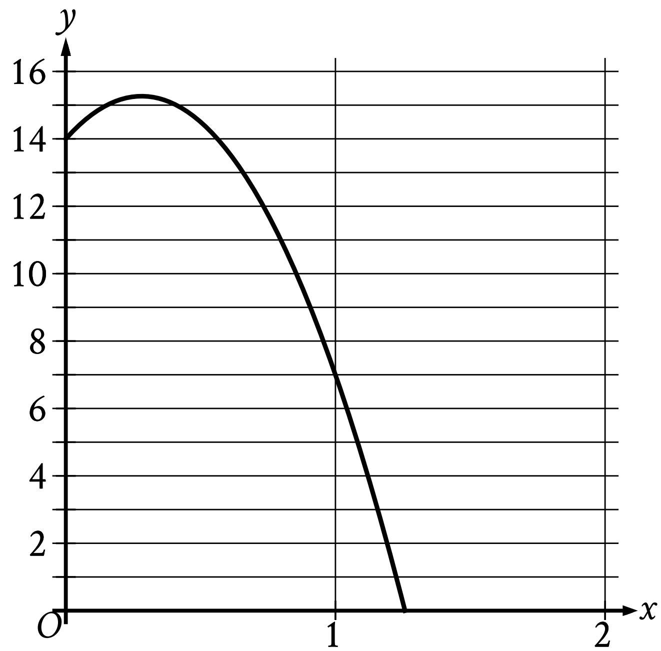
- The parabola opens downward.
- The vertex is at the approximate point (0.3 comma 6.3).
- The parabola passes through the following points:
  - (0 comma 5)
  - approximately (0.3 comma 6.3)
  - approximately (0.6 comma 5)

C.



- The parabola opens downward.
- The vertex is at the approximate point (0.3 comma 10.3).
- The parabola passes through the following points:
  - (0 comma 9)
  - approximately (0.3 comma 10.3)
  - approximately (0.6 comma 9)

D.



- The parabola opens downward.
- The vertex is at the approximate point (0.3 comma 15.3).
- The parabola passes through the following points:
  - (0 comma 14)
  - approximately (0.3 comma 15.3)
  - approximately (0.6 comma 14)

Q7

ADVANCED MATH

The function f is defined by $f x = \frac{1}{6x}$. What is the value of $f x$ when $x = 3$?

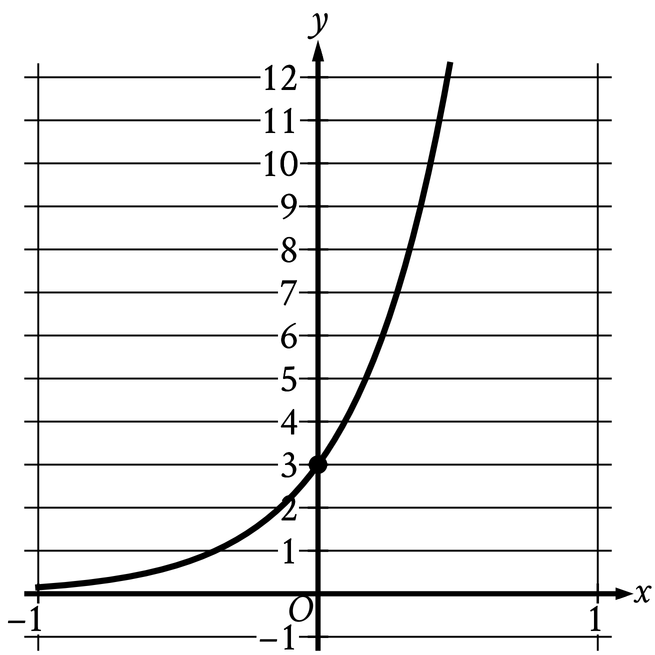
A. $\frac{1}{3}$

B. $\frac{1}{6}$

C. $\frac{1}{9}$

D. $\frac{1}{18}$

Q8 GRID-IN ADVANCED MATH



- Moving from left to right:
  - The curve passes from quadrant 2 to quadrant 1.
  - In quadrant 2, the curve trends up sharply to the point (0 comma 3).
  - In quadrant 1, the curve trends up sharply.
- The curve passes through the following points:
  - approximately (negative 1 comma 0.2)
  - (0 comma 3)

The graph of the exponential function f is shown, where $y = fx$. The y-intercept of the graph is $0, y$. What is the value of y?

STUDENT-PRODUCED RESPONSE

1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4
5 5 5 5
6 6 6 6
7 7 7 7
8 8 8 8
9 9 9 9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9 GRID-IN ADVANCED MATH

The function h is defined by $hx = \frac{8}{5x+6}$. What is the value of $h2$?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q10 ADVANCED MATH

A ball is dropped from an initial height of 22 feet and bounces off the ground repeatedly. The function h estimates that the maximum height reached after each time the ball hits the ground is 85% of the maximum height reached after the previous time the ball hit the ground. Which equation defines h, where $h(n)$ is the estimated maximum height of the ball after it has hit the ground n times and n is a whole number greater than 1 and less than 10?

A. $h(n) = 22(0.22)^n$

B. $h(n) = 22(0.85)^n$

C. $h(n) = 850.22^n$

D. $h(n) = 85(0.85)^n$